

Spotify Music Analytics Dashboard – Power BI Project Report

Project Overview

The Spotify Music Analytics Dashboard is an interactive Power BI project developed to analyze Spotify music data and generate meaningful insights related to artist performance, song popularity, music release trends, and content distribution.

The dashboard transforms raw Spotify data into actionable insights through data cleaning, data modeling, DAX calculations, and interactive visualizations. It provides users with a comprehensive understanding of music trends and listener preferences while demonstrating practical business intelligence and data analytics skills.

Objectives

The primary objectives of this project are:

- Analyze Spotify music trends over time.
 - Identify top-performing artists and songs.
 - Measure overall music popularity.
 - Understand track characteristics such as energy and danceability.
 - Compare explicit and non-explicit content.
 - Create an interactive dashboard for effective data exploration and decision-making.
-

Dataset Description

The dataset contains Spotify track information with the following attributes:

Column	Description
id	Unique Track Identifier
name	Song Name
artists	Artist Name
year	Release Year
popularity	Popularity Score (0–100)
danceability	Danceability Score
energy	Energy Score
valence	Musical Positivity

Column	Description
acousticness	Acousticness Score
speechiness	Speech Content Score
instrumentalness	Instrumentalness Score
liveness	Live Performance Probability
tempo	Tempo (BPM)
explicit	Explicit Content Indicator
duration_min	Song Duration (Minutes)

Data Cleaning Process

The dataset was cleaned and transformed using Power Query before analysis.

Data Cleaning Steps

- Removed unnecessary columns.
- Handled missing and inconsistent values.
- Corrected data types.
- Renamed columns for better readability.
- Converted song duration from milliseconds to minutes.
- Verified numerical fields for accurate calculations and aggregation.
- Optimized the dataset structure for dashboard development.

Final Dataset Structure

The final dataset contains 15 optimized columns suitable for analytical reporting and visualization.

Data Modeling

A single-table analytical model was implemented due to the clean and structured nature of the dataset.

Since all required attributes were available within one table, no additional relationships or bridge tables were required.

DAX Measures

Total Tracks

```
Total Songs =  
COUNT(id)
```

Average Popularity

```
Avg Popularity =  
AVERAGE(popularity)
```

Average Energy

```
Avg Energy =  
AVERAGE(energy)
```

Average Danceability

```
Avg Danceability =  
AVERAGE(danceability)
```

Average Tempo

```
Avg Tempo =  
AVERAGE(tempo)
```

Songs Count

```
Songs Count =  
COUNT(id)
```

Dashboard Components

KPI Metrics

The dashboard contains five key performance indicators that provide a quick overview of the dataset:

1. Total Tracks
2. Average Popularity

3. Average Energy
4. Average Danceability
5. Average Tempo

These metrics help users quickly understand the overall characteristics of the Spotify music catalog.

Top 10 Artists by Average Popularity

Purpose

To identify artists with the highest average popularity scores.

Insight

This analysis highlights artists that consistently maintain strong audience engagement and popularity across their tracks.

Popularity Trend Over the Years

Purpose

To analyze how song popularity has evolved over time.

Insight

The trend reveals changes in listener preferences and helps identify periods of increasing or decreasing popularity.

Songs Released by Year

Purpose

To measure the number of tracks released across different years.

Insight

The analysis shows how music production has expanded over time and highlights periods of significant growth.

Explicit vs Non-Explicit Tracks

Purpose

To compare the distribution of explicit and non-explicit songs.

Insight

This analysis provides an overview of content composition and reveals the prevalence of explicit content in the Spotify catalog.

Top 10 Most Popular Songs

Purpose

To identify the highest-performing songs based on popularity scores.

Insight

This visualization highlights tracks that have achieved exceptional audience engagement and recognition.

Top 10 Artists by Number of Tracks

Purpose

To identify artists with the largest number of tracks in the dataset.

Insight

The analysis measures artist productivity and highlights contributors with extensive music catalogs.

Filters and Interactivity

The dashboard includes interactive filters that allow users to customize their analysis.

Available Filters

- Year Filter
- Artist Filter
- Explicit Content Filter

These filters enable dynamic exploration of the dataset and improve user interaction with the dashboard.

Key Insights

Artist Performance

A relatively small number of artists dominate popularity rankings and maintain consistently high audience engagement.

Growth in Music Production

The volume of music releases has increased substantially over time, reflecting the growth of streaming platforms and digital music consumption.

Song Popularity Distribution

Only a limited number of songs achieve exceptionally high popularity scores, demonstrating the competitive nature of the music industry.

Content Composition

Both explicit and non-explicit tracks contribute significantly to the overall music catalog, providing a diverse range of content for listeners.

Evolving Listener Preferences

Popularity trends vary across years, indicating changes in audience interests, music styles, and listening behavior.

Skills Demonstrated

This project demonstrates practical expertise in:

- Data Cleaning
- Data Transformation
- Power Query
- DAX Calculations
- Data Modeling
- Data Visualization
- Dashboard Development
- Business Intelligence Reporting
- Analytical Thinking
- Data Storytelling

Tools Used

- Microsoft Power BI Desktop
- Power Query

- DAX
 - Microsoft Excel / CSV Dataset
-

Conclusion

The Spotify Music Analytics Dashboard successfully converts raw Spotify data into meaningful business insights through interactive reporting and analysis. By examining artist performance, song popularity, release trends, and content distribution, the dashboard provides a comprehensive understanding of Spotify music data. This project showcases end-to-end data analytics capabilities and demonstrates the effective use of Power BI for transforming complex datasets into actionable insights.